



Chapter 3: Determinants

Question 1:

Compute the determinants by cofactor expansions. At each step, choose a row or column that involves the least amount of computation.

a)

$$\begin{vmatrix} 6 & 0 & 0 & 5 \\ 1 & 7 & 2 & -5 \\ 2 & 0 & 0 & 0 \\ 8 & 3 & 1 & 8 \end{vmatrix}$$

b)

$$\begin{vmatrix} 1 & -2 & 5 & 2 \\ 0 & 0 & 3 & 0 \\ 2 & -6 & -7 & 5 \\ 5 & 0 & 4 & 4 \end{vmatrix}$$

First expand along the third row, then expand along the first row of the remaining matrix:

$$\begin{vmatrix} 6 & 0 & 0 & 5 \\ 1 & 7 & 2 & -5 \\ 2 & 0 & 0 & 0 \\ 8 & 3 & 1 & 8 \end{vmatrix} = (-1)^{3+1} \cdot 2 \begin{vmatrix} 0 & 0 & 5 \\ 7 & 2 & -5 \\ 3 & 1 & 8 \end{vmatrix} = 2 \cdot (-1)^{1+3} \cdot 5 \begin{vmatrix} 7 & 2 \\ 3 & 1 \end{vmatrix} = 10(1) = 10$$

First expand along the second row, then expand along either the third row or the second column of the remaining matrix.

$$\begin{vmatrix} 1 & -2 & 5 & 2 \\ 0 & 0 & 3 & 0 \\ 2 & -6 & -7 & 5 \\ 5 & 0 & 4 & 4 \end{vmatrix} = (-1)^{2+3} \cdot 3 \begin{vmatrix} 1 & -2 & 2 \\ 2 & -6 & 5 \\ 5 & 0 & 4 \end{vmatrix} \\ = (-3) \left((-1)^{3+1} \cdot 5 \begin{vmatrix} -2 & 2 \\ -6 & 5 \end{vmatrix} + (-1)^{3+3} \cdot 4 \begin{vmatrix} 1 & -2 \\ 2 & -6 \end{vmatrix} \right) = (-3)(5(2) + 4(-2)) = -6$$

Question 2:

Compute the determinant of the following matrix:

$$\begin{vmatrix} 4 & 0 & 0 & 0 \\ 7 & -1 & 0 & 0 \\ 2 & 6 & 3 & 0 \\ 5 & -8 & 4 & -3 \end{vmatrix}$$

It's a triangular matrix and thus the determinant is the product of the terms in the diagonal = $4 * -1 * 3 * -3 = 36$

Question 3:

Combine the methods of row reduction and cofactor expansion to compute the determinants.

a)

$$\begin{vmatrix} 2 & 5 & -3 & -1 \\ 3 & 0 & 1 & -3 \\ -6 & 0 & -4 & 9 \\ 4 & 10 & -4 & -1 \end{vmatrix}$$

b)

$$\begin{vmatrix} -1 & 2 & 3 & 0 \\ 3 & 4 & 3 & 0 \\ 5 & 4 & 6 & 6 \\ 4 & 2 & 4 & 3 \end{vmatrix}$$

First use a row replacement to create zeros in the second column, and then expand down the second column:

$$\begin{vmatrix} 2 & 5 & -3 & -1 \\ 3 & 0 & 1 & -3 \\ -6 & 0 & -4 & 9 \\ 4 & 10 & -4 & -1 \end{vmatrix} = \begin{vmatrix} 2 & 5 & -3 & -1 \\ 3 & 0 & 1 & -3 \\ -6 & 0 & -4 & 9 \\ 0 & 0 & 2 & 1 \end{vmatrix} = -5 \begin{vmatrix} 3 & 1 & -3 \\ -6 & -4 & 9 \\ 0 & 2 & 1 \end{vmatrix}$$

Now use a row replacement to create zeros in the first column, and then expand down the first column:

$$-5 \begin{vmatrix} 3 & 1 & -3 \\ -6 & -4 & 9 \\ 0 & 2 & 1 \end{vmatrix} = -5 \begin{vmatrix} 3 & 1 & -3 \\ 0 & -2 & 3 \\ 0 & 2 & 1 \end{vmatrix} = (-5)(3) \begin{vmatrix} -2 & 3 \\ 2 & 1 \end{vmatrix} = (-5)(3)(-8) = 120$$

First use a row replacement to create zeros in the fourth column, and then expand down the fourth column:

$$\begin{vmatrix} -1 & 2 & 3 & 0 \\ 3 & 4 & 3 & 0 \\ 5 & 4 & 6 & 6 \\ 4 & 2 & 4 & 3 \end{vmatrix} = \begin{vmatrix} -1 & 2 & 3 & 0 \\ 3 & 4 & 3 & 0 \\ -3 & 0 & -2 & 0 \\ 4 & 2 & 4 & 3 \end{vmatrix} = 3 \begin{vmatrix} -1 & 2 & 3 \\ 3 & 4 & 3 \\ -3 & 0 & -2 \end{vmatrix}$$

Now use a row replacement to create zeros in the first column, and then expand down the first column:

$$3 \begin{vmatrix} -1 & 2 & 3 \\ 3 & 4 & 3 \\ -3 & 0 & -2 \end{vmatrix} = 3 \begin{vmatrix} -1 & 2 & 3 \\ 0 & 10 & 12 \\ 0 & -6 & -11 \end{vmatrix} = 3(-1) \begin{vmatrix} 10 & 12 \\ -6 & -11 \end{vmatrix} = 3(-1)(-38) = 114$$

Question 4:

Use the determinant to check whether the set of vectors is linearly independent:

a) $v_1 = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} 3 \\ 3 \\ 2 \end{bmatrix}$ and $v_3 = \begin{bmatrix} 0 \\ 4 \\ 1 \end{bmatrix}$

b) $v_1 = \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 0 \\ -3 \\ 5 \end{bmatrix}$ and $v_3 = \begin{bmatrix} -1 \\ -2 \\ 3 \end{bmatrix}$

Since $\begin{vmatrix} 2 & 3 & 0 \\ 1 & 3 & 4 \\ 1 & 2 & 1 \end{vmatrix} = -1 \neq 0$, the matrix is invertible.

Since $\begin{vmatrix} 5 & 0 & -1 \\ 1 & -3 & -2 \\ 0 & 5 & 3 \end{vmatrix} = 0$, the matrix is not invertible.

Question 5:

Find the determinants where:

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = 7.$$

a)

$$\begin{vmatrix} a & b & c \\ d & e & f \\ 3g & 3h & 3i \end{vmatrix}$$

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b)

$$\begin{vmatrix} a & b & c \\ 5d & 5e & 5f \\ g & h & i \end{vmatrix}$$

35

c)
det of

7

d)

$$\begin{vmatrix} d & e & f \\ a & b & c \\ g & h & i \end{vmatrix}$$

-7

Question 6:

Let A and B be 3 x 3 matrices, with $\det A = 4$ and $\det B = -3$. Use properties of determinants to compute.

a) $\det AB$

b) $\det 5A$

c) $\det B^T$

d) $\det A^T B A$

e) $\det B^{-1} A B$

a) -12

b) $4 \cdot 5 \cdot 5 \cdot 5 = 500$

c) -3

d) $4 \cdot -3 \cdot 4 = -48$

e) 4

Question 7:

Use Cramer's rule to compute the solutions of the systems in:

$$\text{a) } \begin{cases} 5x_1 + 7x_2 = 3 \\ 2x_1 + 4x_2 = 1 \end{cases}$$

$$\text{b) } \begin{cases} 4x_1 + x_2 = 6 \\ 5x_1 + 2x_2 = 7 \end{cases}$$

$$\text{c) } \begin{cases} 3x_1 - 2x_2 = 7 \\ -5x_1 + 6x_2 = -5 \end{cases}$$

$$d) \begin{cases} -5x_1 + 3x_2 = 9 \\ 3x_1 - x_2 = -5 \end{cases}$$

The system is equivalent to $A\mathbf{x} = \mathbf{b}$, where $A = \begin{bmatrix} 5 & 7 \\ 2 & 4 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$. We compute

$$A_1(\mathbf{b}) = \begin{bmatrix} 3 & 7 \\ 1 & 4 \end{bmatrix}, A_2(\mathbf{b}) = \begin{bmatrix} 5 & 3 \\ 2 & 1 \end{bmatrix}, \det A = 6, \det A_1(\mathbf{b}) = 5, \det A_2(\mathbf{b}) = -1,$$

$$x_1 = \frac{\det A_1(\mathbf{b})}{\det A} = \frac{5}{6}, x_2 = \frac{\det A_2(\mathbf{b})}{\det A} = -\frac{1}{6}.$$

The system is equivalent to $A\mathbf{x} = \mathbf{b}$, where $A = \begin{bmatrix} 4 & 1 \\ 5 & 2 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 6 \\ 7 \end{bmatrix}$. We compute

$$A_1(\mathbf{b}) = \begin{bmatrix} 6 & 1 \\ 7 & 2 \end{bmatrix}, A_2(\mathbf{b}) = \begin{bmatrix} 4 & 6 \\ 5 & 7 \end{bmatrix}, \det A = 3, \det A_1(\mathbf{b}) = 5, \det A_2(\mathbf{b}) = -2,$$

$$x_1 = \frac{\det A_1(\mathbf{b})}{\det A} = \frac{5}{3}, x_2 = \frac{\det A_2(\mathbf{b})}{\det A} = -\frac{2}{3}.$$

The system is equivalent to $A\mathbf{x} = \mathbf{b}$, where $A = \begin{bmatrix} 3 & -2 \\ -5 & 6 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 7 \\ -5 \end{bmatrix}$. We compute

$$A_1(\mathbf{b}) = \begin{bmatrix} 7 & -2 \\ -5 & 6 \end{bmatrix}, A_2(\mathbf{b}) = \begin{bmatrix} 3 & 7 \\ -5 & -5 \end{bmatrix}, \det A = 8, \det A_1(\mathbf{b}) = 32, \det A_2(\mathbf{b}) = 20,$$

$$x_1 = \frac{\det A_1(\mathbf{b})}{\det A} = \frac{32}{8} = 4, x_2 = \frac{\det A_2(\mathbf{b})}{\det A} = \frac{20}{8} = \frac{5}{2}.$$

The system is equivalent to $A\mathbf{x} = \mathbf{b}$, where $A = \begin{bmatrix} -5 & 3 \\ 3 & -1 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 9 \\ -5 \end{bmatrix}$. We compute

$$A_1(\mathbf{b}) = \begin{bmatrix} 9 & 3 \\ -5 & -1 \end{bmatrix}, A_2(\mathbf{b}) = \begin{bmatrix} -5 & 9 \\ 3 & -5 \end{bmatrix}, \det A = -4, \det A_1(\mathbf{b}) = 6, \det A_2(\mathbf{b}) = -2,$$

$$x_1 = \frac{\det A_1(\mathbf{b})}{\det A} = \frac{6}{-4} = -\frac{3}{2}, x_2 = \frac{\det A_2(\mathbf{b})}{\det A} = \frac{-2}{-4} = \frac{1}{2}.$$

Question 8:

Compute the adjugate of the given matrix and then use the theorem 8 to give the inverse of the matrix:

$$\text{a) } \begin{bmatrix} 0 & -2 & -1 \\ 3 & 0 & 0 \\ -1 & 1 & 1 \end{bmatrix}$$

$$\text{b) } \begin{bmatrix} 1 & 1 & 3 \\ 2 & -2 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Since $\det A = 3$ and the cofactors of the given matrix are

$$C_{11} = \begin{vmatrix} 0 & 0 \\ 1 & 1 \end{vmatrix} = 0, \quad C_{12} = -\begin{vmatrix} 3 & 0 \\ -1 & 1 \end{vmatrix} = -3, \quad C_{13} = \begin{vmatrix} 3 & 0 \\ -1 & 1 \end{vmatrix} = 3,$$

$$C_{21} = -\begin{vmatrix} -2 & -1 \\ 1 & 1 \end{vmatrix} = 1, \quad C_{22} = \begin{vmatrix} 0 & -1 \\ -1 & 1 \end{vmatrix} = -1, \quad C_{23} = -\begin{vmatrix} 0 & -2 \\ -1 & 1 \end{vmatrix} = 2,$$

$$C_{31} = \begin{vmatrix} -2 & -1 \\ 0 & 0 \end{vmatrix} = 0, \quad C_{32} = -\begin{vmatrix} 0 & -1 \\ 3 & 0 \end{vmatrix} = -3, \quad C_{33} = \begin{vmatrix} 0 & -2 \\ 3 & 0 \end{vmatrix} = 6,$$

$$\text{adj } A = \begin{bmatrix} 0 & 1 & 0 \\ -3 & -1 & -3 \\ 3 & 2 & 6 \end{bmatrix} \text{ and } A^{-1} = \frac{1}{\det A} \text{adj } A = \begin{bmatrix} 0 & 1/3 & 0 \\ -1 & -1/3 & -1 \\ 1 & 2/3 & 2 \end{bmatrix}.$$

Since $\det A = 5$ and the cofactors of the given matrix are

$$C_{11} = \begin{vmatrix} -2 & 1 \\ 1 & 0 \end{vmatrix} = -1, \quad C_{12} = -\begin{vmatrix} 2 & 1 \\ 0 & 0 \end{vmatrix} = 0, \quad C_{13} = \begin{vmatrix} 2 & -2 \\ 0 & 1 \end{vmatrix} = 2,$$

$$C_{21} = -\begin{vmatrix} 1 & 3 \\ 1 & 0 \end{vmatrix} = 3, \quad C_{22} = \begin{vmatrix} 1 & 3 \\ 0 & 0 \end{vmatrix} = 0, \quad C_{23} = -\begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} = -1,$$

$$C_{31} = \begin{vmatrix} 1 & 3 \\ -2 & 1 \end{vmatrix} = 7, \quad C_{32} = -\begin{vmatrix} 1 & 3 \\ 2 & 1 \end{vmatrix} = 5, \quad C_{33} = \begin{vmatrix} 1 & 1 \\ 2 & -2 \end{vmatrix} = -4,$$

$$\text{adj } A = \begin{bmatrix} -1 & 3 & 7 \\ 0 & 0 & 5 \\ 2 & -1 & -4 \end{bmatrix} \text{ and } A^{-1} = \frac{1}{\det A} \text{adj } A = \begin{bmatrix} -1/5 & 3/5 & 7/5 \\ 0 & 0 & 1 \\ 2/5 & -1/5 & -4/5 \end{bmatrix}.$$

Question 9:

Find the volume of the parallelepiped with one vertex at the origin and adjacent vertices at $(1,0,-2)$, $(1,2,4)$, and $(7,1,0)$.

The parallelepiped is determined by the columns of $A = \begin{bmatrix} 1 & 1 & 7 \\ 0 & 2 & 1 \\ -2 & 4 & 0 \end{bmatrix}$, so the volume of the parallelepiped is $|\det A| = |22| = 22$.

Question 10:

Let S be the parallelogram determined by the vectors $b_1 = \begin{bmatrix} -3 \\ 2 \end{bmatrix}$ and $b_2 = \begin{bmatrix} -3 \\ 6 \end{bmatrix}$ and $b_3 = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$ and let $A = \begin{bmatrix} 6 & -2 \\ -3 & 2 \end{bmatrix}$. Compute the area of the image of S under the mapping $x \rightarrow Ax$.

Apply a translation of $(1,1)$ to get the origin and then use the method below:

Since the parallelogram S is determined by the columns of $\begin{bmatrix} -2 & -2 \\ 3 & 7 \end{bmatrix}$, the area of S is

$\left| \det \begin{bmatrix} -2 & -2 \\ 3 & 7 \end{bmatrix} \right| = |-8| = 8$. The matrix A has $\det A = \begin{vmatrix} 6 & -2 \\ -3 & 2 \end{vmatrix} = 6$. By Theorem 10, the area of $T(S)$ is $|\det A| \{\text{area of } S\} = 6 \cdot 8 = 48$.